

FC01 compact two-nozzle filling cell

Siemens-authoritative controls and bounded NVIDIA quality architecture

FICTIONAL ENGINEERING PROJECT - NOT FOR CONSTRUCTION

CONCEPTUAL SAFETY ARCHITECTURE - REQUIRES PROJECT-SPECIFIC RISK ASSESSMENT, DESIGN, VERIFICATION AND VALIDATION BY A QUALIFIED MACHINERY-SAFETY ENGINEER.
NO PERFORMANCE LEVEL, SIL, CATEGORY, CE CONFORMITY OR REGULATORY COMPLIANCE IS CLAIMED.

Release status	What is implemented	What remains blocked
NVIDIA IMPLEMENTATION-READINESS RELEASE CANDIDATE	Revision-F canonical model; Git-object byte authority; 47-node immutable PLC/AI contract; production-shaped edge and dataset tools; 32 process scenarios plus 28 structured behavioral vision fault injections; inherited native CAD/QET evidence.	TIA/WinCC compile/archive; Startdrive; PLCSIM; final site electrical inputs; real dataset/model/target runtime; physical FAT/SAT and qualified safety work.

This report is an evidence index. It is not proof of construction readiness, functional safety, native compile, FAT, SAT, electrical test, physical commissioning or AI performance.

Integrated architecture and deterministic ownership

The PLC owns all machine sequence, interlocks, timeouts, product disposition and transfer permission. THE NVIDIA VISION SUBSYSTEM IS NON-SAFETY-RELATED AND MUST NOT BE USED AS THE SOLE MEANS OF PERSONNEL PROTECTION, SAFE STOP, GUARD MONITORING OR HAZARDOUS-MOTION CONTROL.

Layer	Owner	Deterministic responsibility	Failure behavior
Field / power	Electrical	Identified device, cable/core, terminal and I/O channel	Fail-closed valves; external safety removes hazardous energy
Equipment control	S7-1500 FB instances	Gate, clamp, two fill channels, conveyor/pump VFD, capper	Module timeout/contradiction fault; outputs safe
Machine coordinator	S7-1500	Legal states, accepted command edges, no automatic restart	HOLDING, CONTROLLED_STOPPING or FAULTED
Operator interface	MTP700 Unified	Role-checked handshake; feedback and disabled reason	No direct physical-output writes
Quality vision	NVIDIA edge service	Securely acquire request and publish one correlated immutable result	Malformed/stale/low-confidence/fault -> quality hold
Regression	Python + asyncua	Sequence fault injection and encrypted real server/client transport	Design evidence only; never PLCSIM or production-endpoint proof

Machine states: UNINITIALIZED, INITIALIZING, STOPPED, READY, AUTOMATIC, MANUAL_SETUP, HOLDING, CONTROLLED_STOPPING, FAULTED and RECOVERY_RESET. Reset returns to STOPPED and never initiates motion.

Selected Siemens and edge hardware baseline

Function	Selected item	Order number / status	Verification boundary
PLC	SIMATIC S7-1500 CPU 1511-1 PN	6ES7511-1AL03-0AB0	Current official lifecycle checked; V20 device catalog/compile open
Pulse counting	TM Count 2x24V	6ES7550-1AA01-0AB0	Two independent 24 V pulse channels; TIA configuration open
DI / DO / AI	DI 32 HF; DQ 32 HF; AI 8xU/I HF	6ES7521-1BL00-0AB0; 6ES7522-1BL01-0AB0; 6ES7531-7NF00-0AB0	Front connectors, wiring and catalog compatibility open
HMI	MTP700 Unified Comfort	6AV2128-3GB06-0AX1	WinCC V20 compile/licence open
Drives	Two SINAMICS G120C PN 0.75 kW	6SL3210-1KE12-3UF2	Ratings provisional; Startdrive absent
Network	SCALANCE XC208 managed plus S615 firewall	6GK5208-0BA00-2AC2; 6GK5615-0AA01-2AA2	VLAN/firewall architecture selected; native policy and site approval open
Vision	Jetson Orin NX 16GB on industrial carrier	Carrier/integration SKU TBD	Carrier, thermals, EMC, camera and real dataset open

Authoritative engineering target: TIA Portal V20, executable/product version 2000.0.9501.1. Installed STEP 7 and WinCC V20 components do not prove usable licence entitlement. Current identity is not authorized for TIA Openness. Startdrive and PLCSIM are absent.

PLC-AI v3 acceptance contract

Transport selection: encrypted OPC UA across a routed quality zone. The PLC publishes one immutable request and holds its request until coherent capture acknowledgement or a terminal result. Results are accepted only for the current session/ID with fresh heartbeat, healthy service/camera/model, approved model/dataset/calibration identities, allowed disposition and reasons, sufficient confidence, valid timing and a non-contradictory payload.

PLC-owned nodes (13)	NVIDIA-owned nodes (34)	Acceptance rule
Enable, trigger, inspection/session identity, recipe/count/fill target, PLC heartbeat, exact result ACK, maintenance mode and expected dataset/calibration IDs	Ready/busy/capture/state/health; immutable quality payload; disposition/reasons/confidence; model/dataset/calibration identities; diagnostic timestamps/timing; edge heartbeat/diagnostics/queue	Payload publishes before RESULT_VALID and remains immutable until exact ACK. PLC-derived monotonic result age is authoritative; edge UTC is diagnostic. Any stale, malformed, unhealthy, uncertain or contradictory condition enters HOLD/FAULT.

The adapter requires Basic256Sha256 SignAndEncrypt, X.509 application/user identity, controlled trust/CRL stores and an exact 47-node typed map. Local certificate-backed asyncua tests use a synthetic endpoint. No model is delivered. No training, ONNX export, TensorRT engine, DeepStream execution, accuracy, confusion matrix, false-accept/reject analysis or production latency is claimed. The RTX 5060 Laptop GPU does not substitute for the absent CUDA/TAO/DeepStream target stack or representative dataset.

NVIDIA implementation-readiness engineering

Work package	Revision-F controlled result	Open acceptance boundary
Acquisition/runtime	Exactly-once recorded-image source, bounded preprocessing, production-prohibited mock backend and hash-gated ONNX adapter boundary	Physical camera/lens/light, actual tensor contract and target runtime
Dataset/evaluation	Versioned annotation/manifest schemas, file hashes, lot/session split controls, duplicate/leakage checks and FAR/FRR evaluation math	Representative independently labeled multi-lot data, approved thresholds and model-change approval
Platform policy	DeepStream 9.1 and TAO 7.0.1 tracked; generic TensorRT 11.1 is not treated as a Jetson compatibility lock	Selected JetPack/carrier image, CUDA/TensorRT/DeepStream/TAO execution and deployment validation
Electrical delta	Separate provisional 24 V branches: 60 W edge, 12 W camera, 20 W lighting and 8 W service/network; X200 and switch-port reservations	Selected devices, inrush, protection, cable, voltage-drop, thermal, EMC and native schematic/CAD incorporation
Service lifecycle	Schema-controlled configuration, structured diagnostics, durable event-store boundary, health/metrics, rollback and troubleshooting procedures	Site PKI/time/retention/privacy policy, production endpoint and operational acceptance

Synthetic fixtures and local timing harnesses prove software plumbing only. They cannot authorize production READY, establish optical feasibility, or support any accuracy, false-accept, false-reject, throughput or latency claim.

Validation evidence and release blockers

Gate	Evidence	Status
Canonical consistency	551 engineering-validator checks plus 96 dedicated Revision-F checks cover generator parity, schedules, ownership, edge behavioral fault injection, safety boundaries and prohibited artifacts. The standalone electrical delta is not incorporated into every authority.	PARTIAL
Deterministic regression	99 Siemens/simulator/source/native-contract + 86 edge-service + 17 interface-harness tests; 32 process and 28 structured behavioral fault-injection cases	SOURCE/LOCAL PASS - not PLCSIM/HIL
Workbook	26 summary/schedule sheets; deterministic normalized build, formula-error scan and all-sheet rendered review	PASS after final freeze
QElectroTech	Corrected 26-folio QET BCA022BE0B9F... reopened at the exact hash and exported natively to 26 pages. All pages reviewed; folio 25 has one clipped in-body safety sentence and automatic cross-reference resolution is unsupported	NATIVE REOPEN/EXPORT PASS; OVERALL PARTIAL
Panel CAD	FreeCADCmd reopened 165 objects / 148 controlled solids; STEP solid, IGES bounded-face and DXF/hole reimports verified	PASS
TIA / WinCC / drives	V20 installed; no native project, compile, cross-reference, archive restore or Startdrive evidence	BLOCKED
PLCSIM	Not installed	BLOCKED
NVIDIA software	86/86 edge tests including 10 certificate-backed local asyncua cases; 47-node map, terminal-before-valid ordering, durable-audit failure interlock and 28-case behavioral injection pass	PARTIAL - production endpoint/native binding open
NVIDIA model/runtime	Production PLC/PKI, representative dataset, trained model and CUDA/TensorRT/DeepStream target stack absent	MODEL/RUNTIME BLOCKED

Gate totals: 2 PASS, 8 PARTIAL, 11 BLOCKED, 1 OPEN. Release use: controlled design review and continuation in qualified native tools only. Read the manifest, acceptance-gate report and open-input register before work.